ANSWER KEY & MARKING SCHEME SET A

Chapter 2: **The Invisible Living World: Beyond Our Naked Eye** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** microorganisms	Seen only with a microscope
2	**(a)** cell membrane	No cell wall in animal cells
3	**(c)** nucleus	Genetic material lies in the nucleoid
4	**(c)** plants and animals	Complex bodies
5	**(b)** function	Nerve cells are long, red blood cells are disc-like
6	**(d)** Lactobacillus	A bacterium
7	**(a)** microalga used as food supplement	Rich in protein
8	**(d)** reproduce only inside a host organism	Need living cells
9	**(c)** a cell wall	Extra covering outside the membrane
10	**(b)** chlorophyll in chloroplasts	Traps sunlight

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Cell membrane (½). Cytoplasm (½). Nucleus (1).	½+½+1
12	Plant cell has cell wall; animal cell does not (1). Plant cell has chloroplasts; animal cell does not (1).	1+1
13	Region holding genetic material without a membrane (1). Bacterial cells (1).	1+1
14	Bacteria (1). Amoeba or yeast (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Neat diagram (1). Labels: cell membrane, cytoplasm, nucleus (2).	1+2
16	Plant: chloroplast (1). Animal: no cell wall (1). Bacterial: nucleoid (1).	1+1+1
17	Sugar solution with yeast in a test tube, balloon on mouth (1). Kept warm for 3–4 hours (1). Balloon inflates due to carbon dioxide from fermentation (1).	1+1+1
	OR: Yeast in baking (1). Lactobacillus in curd (1). Decomposers clean the environment (1).	1+1+1
18	Rhizobium lives in root nodules (1). It fixes nitrogen from air into usable form (1). The plant gets nitrogen and bacteria get shelter and food (1).	1+1+1
	OR: Cow dung and waste are put in a closed tank (1). Microbes break it down producing methane-rich gas (1). Gas is used for cooking and lighting (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Diagram with labels: cell wall, cell membrane, cytoplasm, nucleus, chloroplast, vacuole (2). Cell wall – support (1). Nucleus – control (1). Chloroplast – photosynthesis (1).	2+1+1+1
	OR: Cell wall present in plant and bacteria (1). Chloroplast only in plant (1). Nucleus well-defined in plant and animal (1). Nucleoid in bacteria (1). Size, vacuole or other point (1).	1+1+1+1+1
20	Cell – red blood cell (1). Tissue – muscle tissue (1). Organ – heart (1). Organ system – circulatory system (1). Organism – human being (1).	1+1+1+1+1
	OR: Both tubes have sugar solution (1). Yeast only in B, balloons on both, kept warm (1). Balloon of B inflates, A does not (1). Yeast produced a gas (1). Gas is carbon dioxide by fermentation (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 2: **The Invisible Living World: Beyond Our Naked Eye** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** cell	All living bodies are made of cells
2	**(d)** cell wall	Plants have extra covering
3	**(d)** nucleoid	No membrane-bound nucleus
4	**(d)** cell → tissue → organ → organ system → organism	Small to large
5	**(d)** Yeast	Used in baking
6	**(b)** legumes	Peas, beans, lentils
7	**(c)** cow dung and organic waste	Anaerobic breakdown
8	**(a)** Paramecium	Single-celled animal-like
9	**(d)** yeast produced a gas	Fermentation releases CO_2
10	**(c)** nucleus, cytoplasm and cell membrane	Typical animal cell

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Cell, tissue, organ (1). Organ system, organism (1).	1+1
12	Yeast ferments sugar and releases carbon dioxide (1). This makes bread, idli, dosa and cakes soft and spongy (1).	1+1
13	It converts milk into curd (1). It is also used in fermentation in food industry (1).	1+1
14	It lives in root nodules of legumes (1). It fixes atmospheric nitrogen and increases soil fertility (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Unicellular: one cell body, e.g. Amoeba, bacteria (1½). Multicellular: many cells, e.g. plants, humans (1½).	1½+1½
16	They are very small and light (1). They need little food (1). They can live in air, water, soil, plants and animals (1).	1+1+1
17	Tissue: group of similar cells, e.g. muscle tissue (1½). Organ: tissues working together, e.g. heart (1½).	1½+1½
	OR: Very small, not like other microorganisms (1). Reproduce only inside a host cell (1). Can cause diseases (1).	1+1+1
18	They break dead plants and animals into simple substances (1½). This recycles nutrients and keeps the environment clean (1½).	1½+1½
	OR: Cell membrane, cytoplasm, genetic material (1). Membrane controls entry and exit (1). Cytoplasm holds cell parts (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Yeast makes bread, idli (1). Lactobacillus makes curd (1). Rhizobium fixes nitrogen (1). Decomposers recycle waste (1). Spirulina gives food supplement (1).	1+1+1+1+1
	OR: Bacteria – unicellular, no true nucleus (1½). Fungi – uni or multicellular (1). Protozoa – unicellular animal-like (1). Viruses – reproduce only in a host (1½).	1½+1+1+1½
20	Slurry of cow dung and water is fed in a digester (1). Microbes break it down without oxygen (1). Methane-rich gas collects (1). Advantage: clean fuel (1). Advantage: leftover slurry is used as manure (1).	1+1+1+1+1
	OR: Shape suits job (1). Nerve cells are long and branched to carry signals (1½). Red blood cells are small discs to move easily in vessels (1½). Plant root hair cells are long to absorb water (1).	1+1½+1½+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.